

EXCALIBUR :

Relativistic raytracing for weak & strong lensing analysis

Laurent Magri-Stella^{1,2}, Nareï Lorenzo Martinez^{2,3}, Vincent Reverdy^{2,3}

¹ Université Savoie Mont Blanc

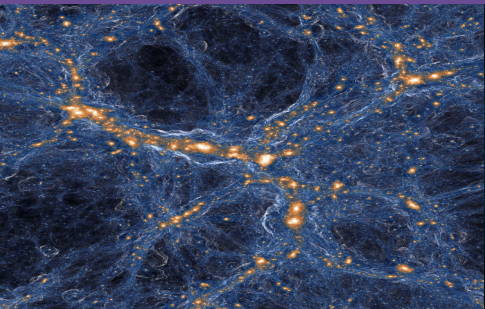
² Laboratoire d'Annecy de Physique des Particules (LAPP), CNRS
9 Chemin de Bellevue, 74940 Annecy-le-Vieux

³ Centre National de la Recherche Scientifique (CNRS)

12.06.2026



Galaxy clusters as a key cosmological probe



The upcoming surveys

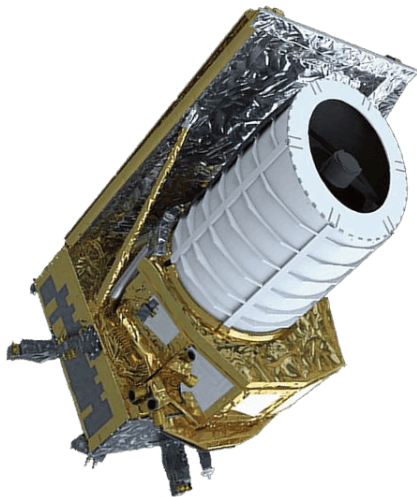
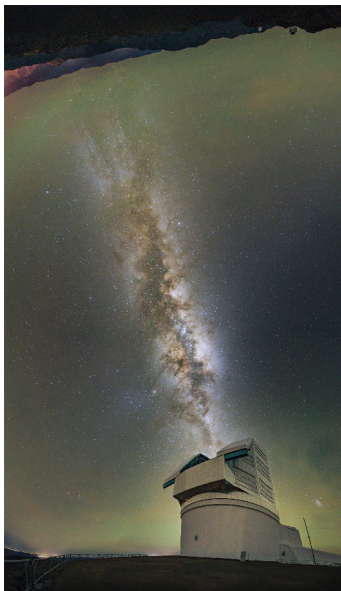
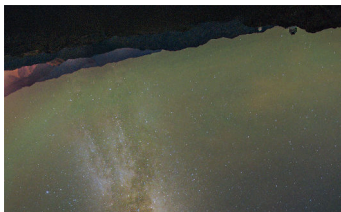


Figure 1: Vera C. Rubin Observatory (LSST) and Euclid satellite.

An important question



How do theoretical uncertainties compare to statistical uncertainties?

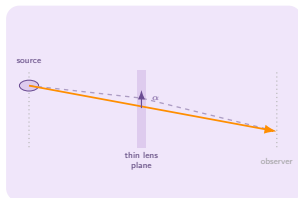


Figure 2: Vera C. Rubin Observatory (LSST) and Euclid satellite.

Towards an answer ?

1

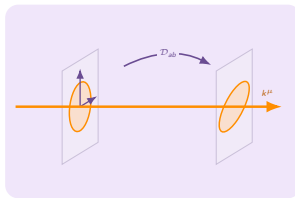
Weak lensing



Current weak-lensing practice and its main approximations.

2

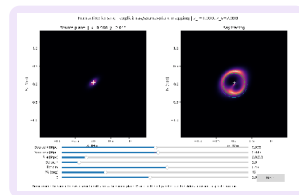
Sachs formalism



Relativistic beam propagation and distortions from curvature.

3

EXCALIBUR



Relativistic raytracing for weak & strong lensing in FLRW space-time.

Weak lensing: a brief overview

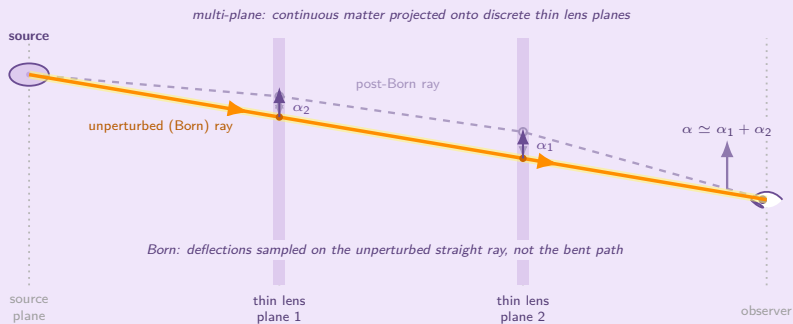


Figure 3: Current approximations in weak lensing frameworks: 1/ Born approximation 2/ thin lens planes. The dashed ray represents the post Born approach, which starts to be more commonly used in simulations, has yet to be implemented in analysis pipelines.

Lens equation and magnification matrix

$$\beta = \theta - \alpha(\theta) \quad \frac{\partial \beta_i}{\partial \theta_j} \approx \delta_{ij} - \frac{\partial \alpha_i}{\partial \theta_j} = A_{ij}$$

Weak lensing: convergence, shear, and flexion

First order approximation of the magnification matrix

$$A = \begin{pmatrix} 1 - \kappa & 0 \\ 0 & 1 - \kappa \end{pmatrix} - \gamma \begin{pmatrix} \cos(2\phi) & \sin(2\phi) \\ \sin(2\phi) & -\cos(2\phi) \end{pmatrix}$$

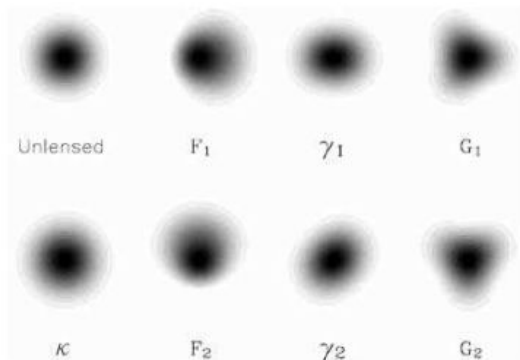


Figure 4: Effects of the different lensing fields on a Gaussian galaxy of radius 1 arcsec. 10% convergence/shear and 0.28 arcsec⁻¹ flexion (which is a very high value for this quantity, chosen only to visualize) are applied. From Bacon et al. 2006

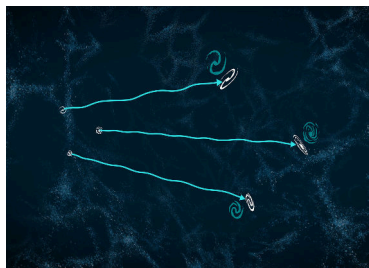
A relativistic description of light propagation

Goal

Describe light propagation directly in space-time, beyond the standard weak-lensing approximations.

Three ingredients

- 1 a background metric describing the geometry;
- 2 the geodesic equation for photon trajectories;
- 3 the Sachs formalism for beam deformations.



Why this drops the approximations

- Born $\rightarrow$ geodesic integrated along the true light path.
- Thin lens $\rightarrow$ continuous fields sampled in 4D.

1/ A background metric describing the geometry

The Friedmann-Lemaitre-Robertson-Walker metric

$$g_{\mu\nu} = \begin{pmatrix} -c^2 & 0 & 0 & 0 \\ 0 & \frac{a^2(t)}{1 - kr^2} & 0 & 0 \\ 0 & 0 & a^2(t)r^2 & 0 \\ 0 & 0 & 0 & a^2(t)r^2 \sin^2 \theta \end{pmatrix}$$

where:

- $a(t)$: scale factor
- k : curvature parameter ($k = 0, +1, -1$)

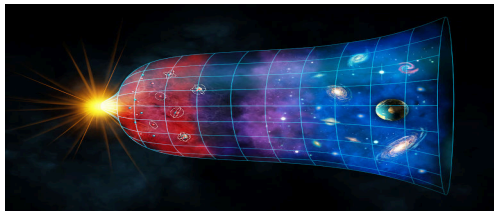


Figure 5: The story of the Universe: from the Big Bang to the present day.
Credit: Natalie Mayer

1/ A perturbed background metric describing the geometry

The perturbed Friedmann-Lemaître-Robertson-Walker metric

$$g_{\mu\nu} = \begin{pmatrix} -c^2(1+2\Phi) & 0 & 0 & 0 \\ 0 & a^2(t)(1-2\Psi) & 0 & 0 \\ 0 & 0 & a^2(t)(1-2\Psi) & 0 \\ 0 & 0 & 0 & a^2(t)(1-2\Psi) \end{pmatrix}$$

in

- cartesian coordinates;
- flat space ($k = 0$).

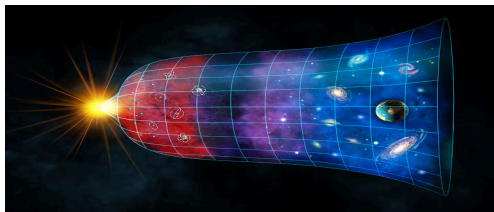
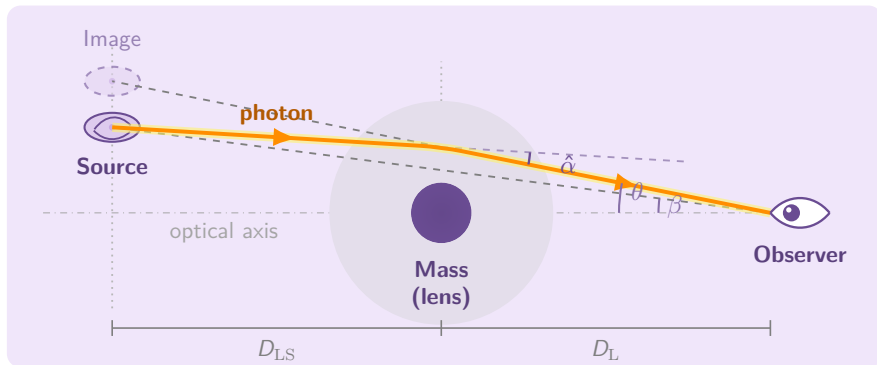


Figure 6: The story of the Universe: from the Big Bang to the present day.

Credit: Natalie Mayer

2/ The geodesic equation for photon trajectories

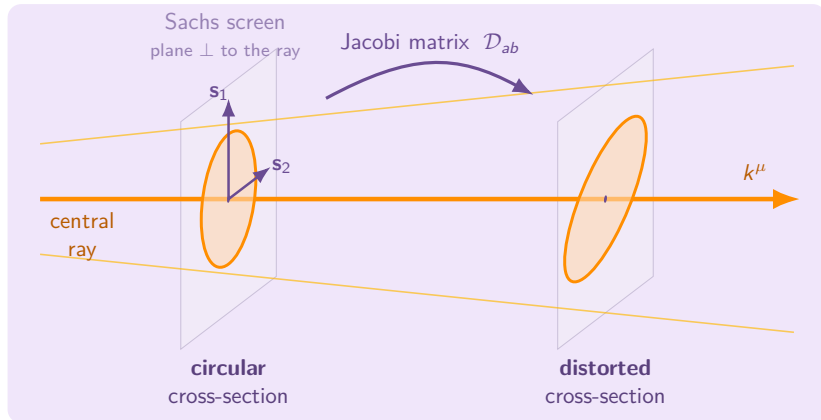


Geodesic equation

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0 \quad g_{\mu\nu} k^\mu k^\nu = 0$$

The metric fixes the Christoffel symbols, which determine how photon trajectories curve through inhomogeneous matter distributions.

3/ The Sachs formalism for beam deformations



Optical tidal matrix

$$\mathcal{R}_{AB} = R_{\mu\nu\rho\sigma} k^\mu s_A^\nu s_B^\rho k^\sigma$$

$$\mathcal{R}_{AB} = -\frac{1}{2} R_{\mu\nu} k^\mu k^\nu \delta_{AB} + C_{\mu\nu\rho\sigma} s_A^\mu k^\nu k^\rho s_B^\sigma$$

Focusing equation and A_{ij}

$$\frac{d^2 \mathcal{D}_{AB}}{dv^2} = \mathcal{R}_{AC} \mathcal{D}_{CB}$$

$$\mathcal{D} = A \bar{\mathcal{D}} + \dots$$

The tensor chain: from the metric to the Jacobi matrix

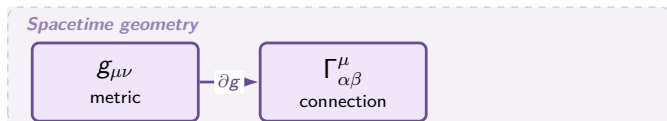
Spacetime geometry

$g_{\mu\nu}$

metric

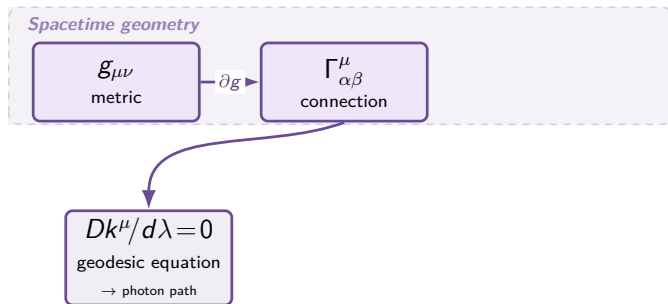
The **metric alone** fixes the whole chain: each tensor follows from derivatives of the previous one.

The tensor chain: from the metric to the Jacobi matrix



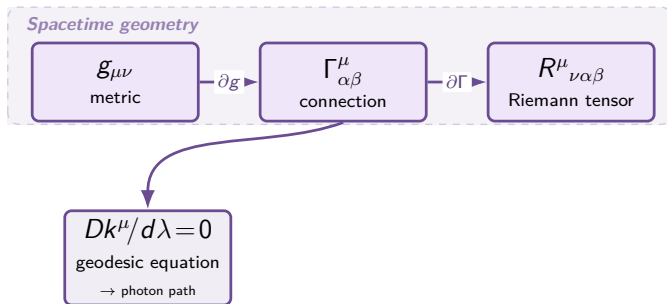
The **metric alone** fixes the whole chain: each tensor follows from derivatives of the previous one.

The tensor chain: from the metric to the Jacobi matrix



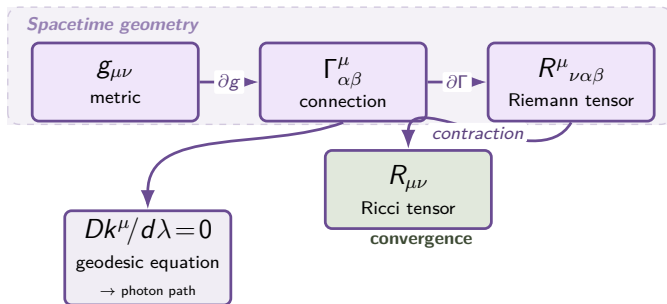
The **metric alone** fixes the whole chain: each tensor follows from derivatives of the previous one.

The tensor chain: from the metric to the Jacobi matrix



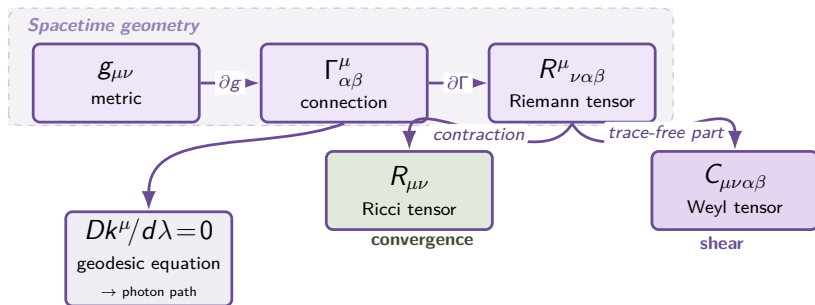
The **metric alone** fixes the whole chain: each tensor follows from derivatives of the previous one.

The tensor chain: from the metric to the Jacobi matrix



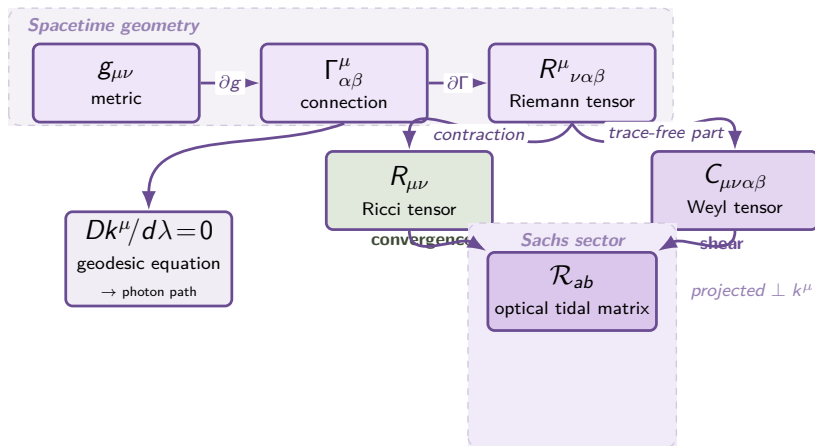
The **metric alone** fixes the whole chain: each tensor follows from derivatives of the previous one.

The tensor chain: from the metric to the Jacobi matrix



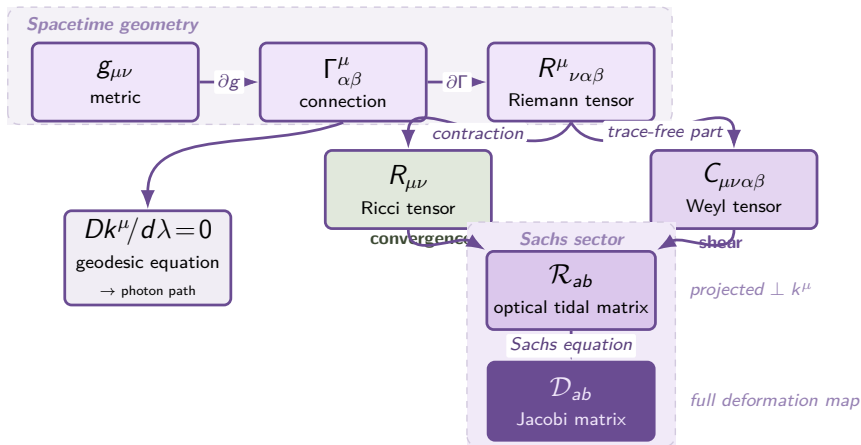
The **metric alone** fixes the whole chain: each tensor follows from derivatives of the previous one.

The tensor chain: from the metric to the Jacobi matrix



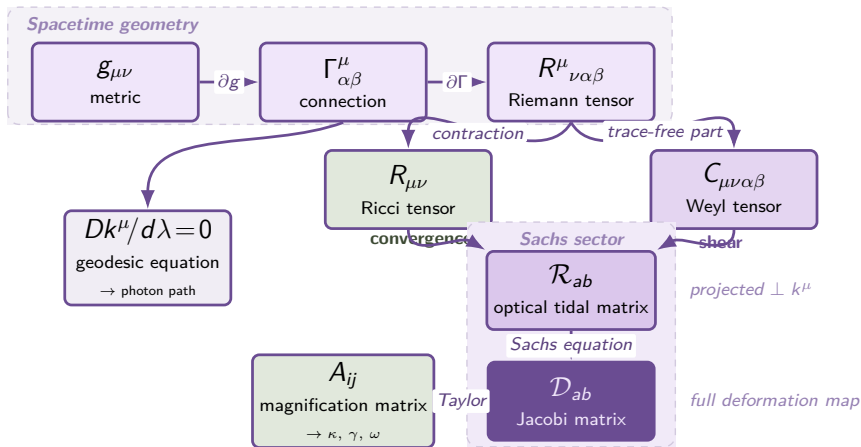
The **metric alone** fixes the whole chain: each tensor follows from derivatives of the previous one.

The tensor chain: from the metric to the Jacobi matrix



The **metric alone** fixes the whole chain: each tensor follows from derivatives of the previous one.

The tensor chain: from the metric to the Jacobi matrix



The **metric alone** fixes the whole chain: each tensor follows from derivatives of the previous one.

Already existing raytracing codes?

Code name	References	Metric	Input	Applications
GEVOLUTION RAYTRACER	J. Adamek 2018	weak-field	gevolution simulations	cosmology, relativistic light cones, redshifts, distances
MAGRATHEA	V. Reverdy 2014, M.A. Breton and V. Reverdy 2021	weak-field	large-scale cosmological simulations	cosmology, relativistic light cones, redshifts, distances, large-scale structures
GYOTO	F. H. Vincent 2011	analytic, 3+1 full GR	analytic models or LORENE code	compact objects
EXCALIBUR	L. Magri-Stella, V. Reverdy, N. Lorenzo-Martinez 2026	analytic, weak-field	analytic models or large-scale structure	cosmology, large-scale structures, gravitational lensing

A gap to fill?

No existing code simultaneously targets weak-field cosmology, large-scale structure, and gravitational lensing in a relativistic framework.

Introducing EXCALIBUR

E X C A L I B U R

EX EXplicit

CA CAlculation (of)

LI LIght

B Beams

U Using

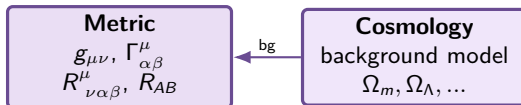
R Relativity

EXCALIBUR aims to be a modular and performant tool to compute light propagation in curved space-times.

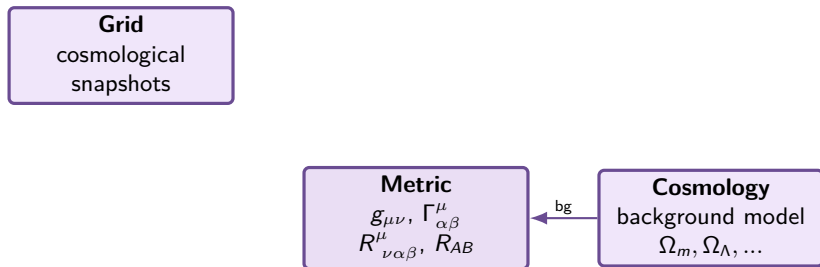
Metric

$$g_{\mu\nu}, \Gamma_{\alpha\beta}^{\mu}$$
$$R^{\mu}_{\nu\alpha\beta}, R_{AB}$$

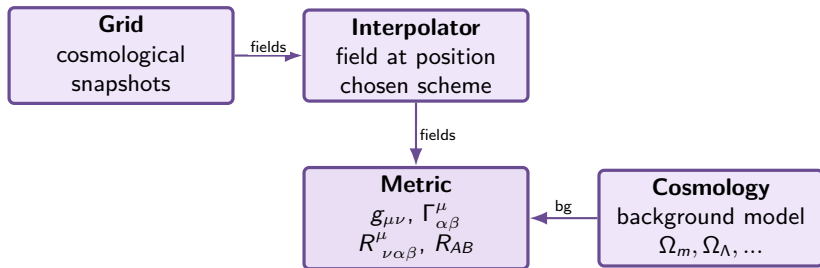
EXCALIBUR: numerical pipeline



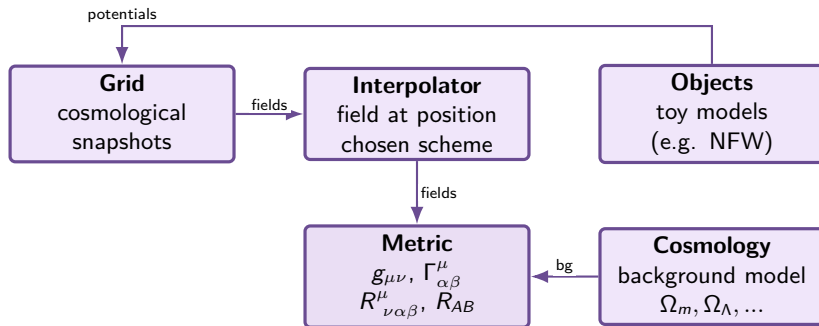
EXCALIBUR: numerical pipeline



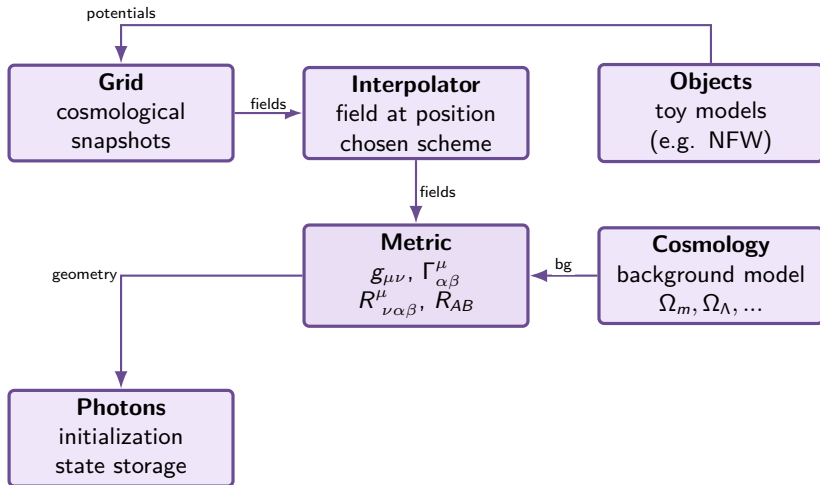
EXCALIBUR: numerical pipeline



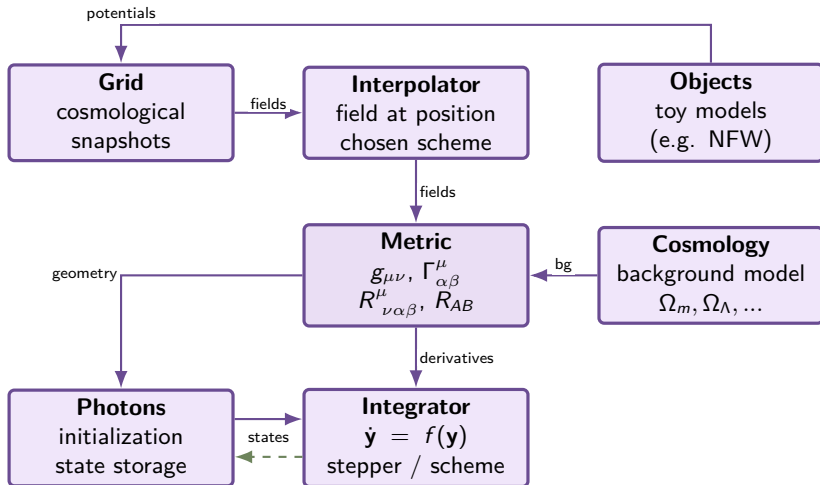
EXCALIBUR: numerical pipeline



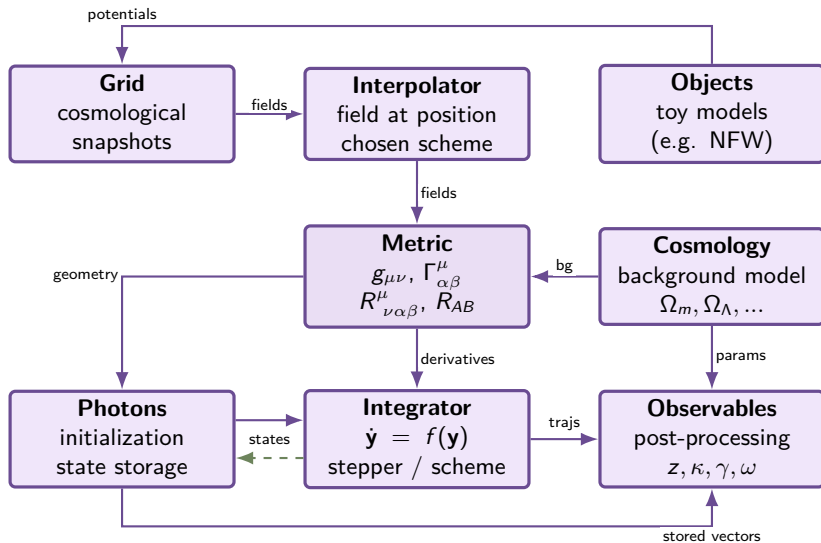
EXCALIBUR: numerical pipeline



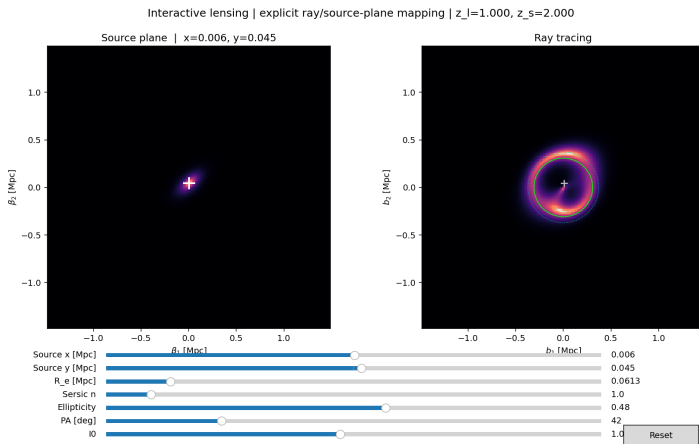
EXCALIBUR: numerical pipeline



EXCALIBUR: numerical pipeline



What EXCALIBUR enables



lmagristella.github.io/random-science/

Code validation

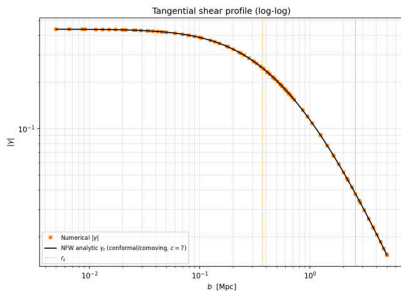


Figure 7: Shear profile of an NFW halo: numerical vs analytical.

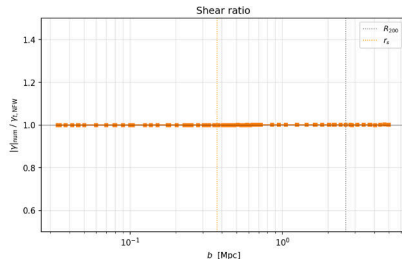
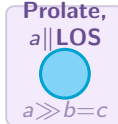
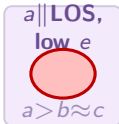
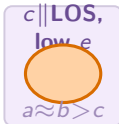
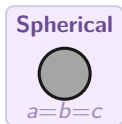
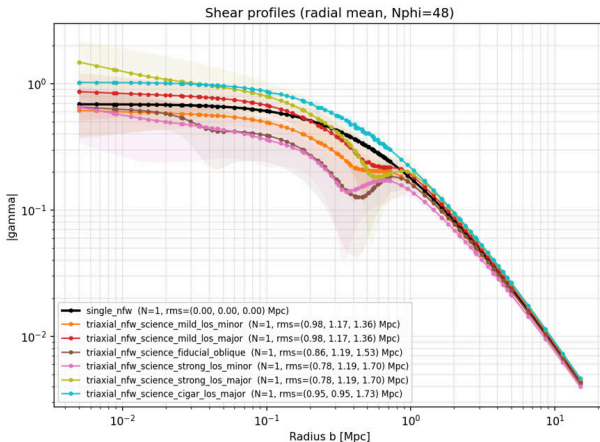


Figure 8: Ratio of numerical to analytical shear profile for an NFW halo.

Validation of lensing quantities

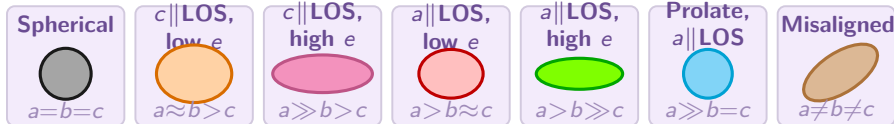
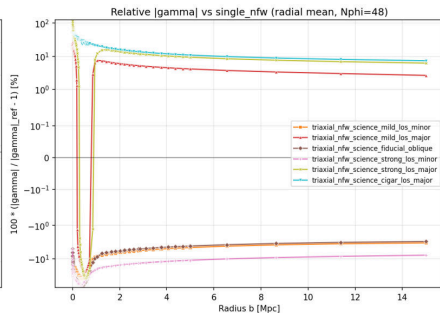
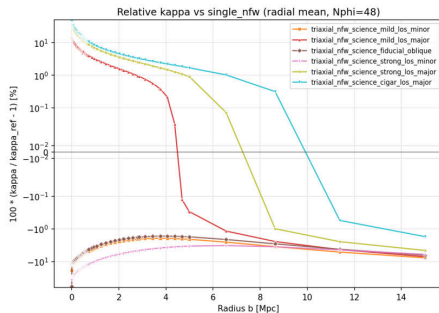
Compare simulated κ and γ profiles of an NFW halo to the analytical NFW predictions.

Current work: model errors on mass and shape



Sky-plane projections ($\perp$ LOS)

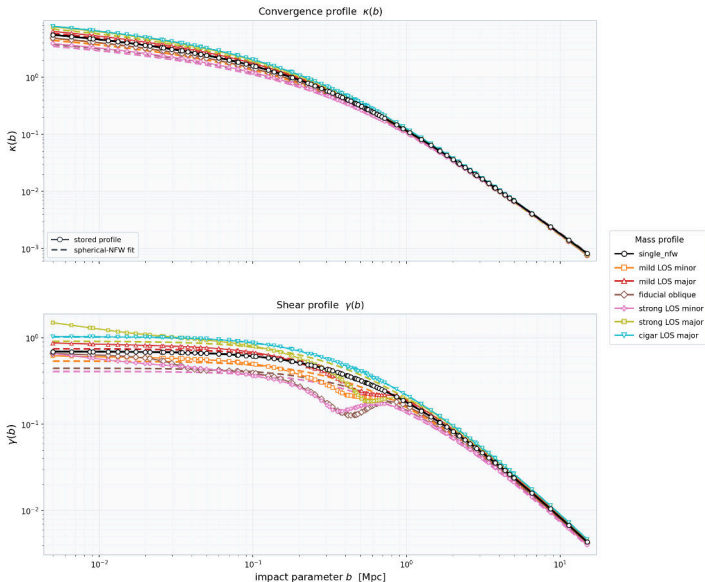
Current work: model errors on mass and shape



Sky-plane projections ($\perp$ LOS)

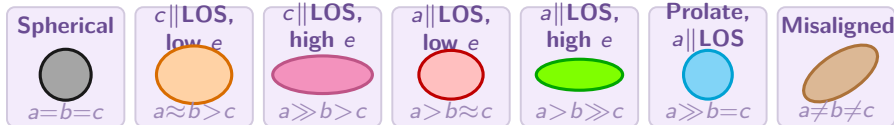
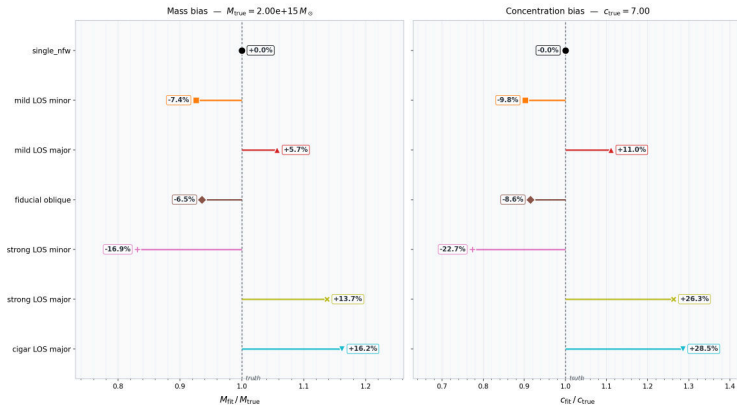
Fitting a spherical NFW model ?

Spherical-NFW fits to equivalent-mass profiles — injected $M_{200} = 2.00e + 15 M_{\odot}$, $c = 7.00$



κ fit biases

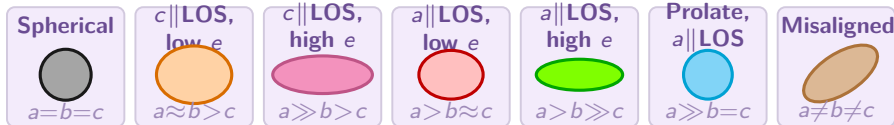
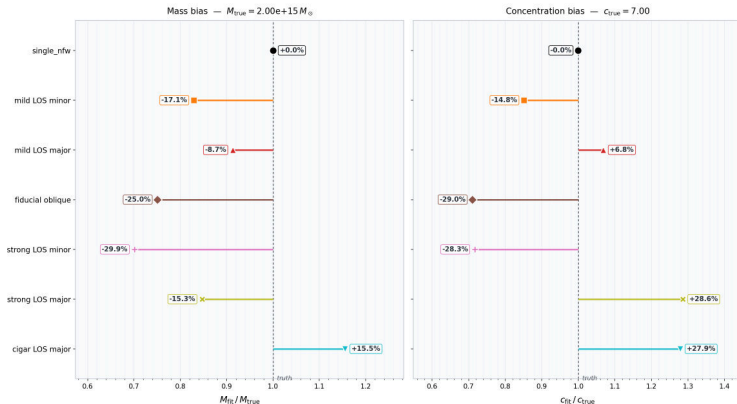
Bias on inferred parameters when fitting a spherical NFW to non-spherical profiles



Sky-plane projections ($\perp$ LOS)

γ fit biases

Bias on inferred parameters when fitting a spherical NFW to non-spherical profiles



Sky-plane projections ($\perp$ LOS)

1

Bias on the cluster mass

5–20%

worst for large ellipticities, $c \parallel \text{LOS}$, and prolate halos

2

Bias on the concentration

5–30%

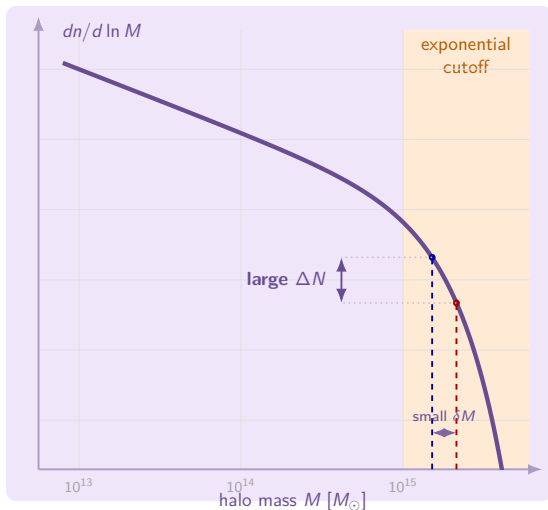
same dependence on the halo geometry

3

Large impact on cosmology

amplified by the exponential cutoff of the halo mass function dN/dM at high mass

The halo mass function amplifies mass biases



Halo mass function

$dn/d \ln M$: comoving number density of halos per unit mass.

Steep cluster-scale tail

At high mass it falls exponentially

$$\frac{dn}{d \ln M} \propto e^{-\delta_c^2/2\sigma^2(M)}$$

Why it matters

A few-% bias on $M \Rightarrow$ **large** bias on cluster counts $\Rightarrow$ biased cosmology (σ_8 , Ω_m).

Consistent with Corless & King (2007)

Same setup

Spherical NFW fits to **triaxial** halos with the odd axis along the LOS

Truth: $M_{200} = 10^{15} M_{\odot}$, $C = 4$

Same trend

- **Prolate** ($Q > 1$) $\Rightarrow$ **overestimated** M and C
- **Oblate** ($Q < 1$) $\Rightarrow$ **underestimated** M and C

Independent confirmation via a purely statistical approach to triaxial halo modeling.

6 *V. L. Corless & L. J. King*

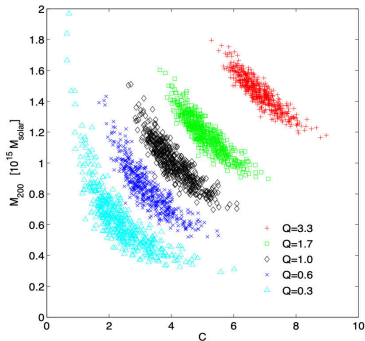
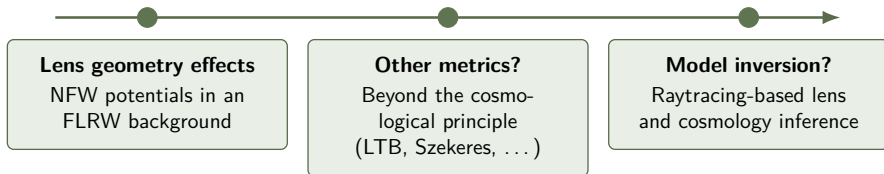


Figure 3. Best-fit concentration C and virial mass M_{200} for a simple NFW model fit to five triaxial lenses with $C = 4$, $M_{200} = 10^{15} M_{\odot}$, circularly symmetric on the sky with odd axis oriented along the line of sight. Halos with axis ratio Q (ratio between odd axis and similar axes) less than one are oblate ($Q = 0.3 \rightarrow a = 0.3$), while those with Q greater than one are prolate ($Q = 3.3 \rightarrow a = b = 0.3$).

Some (yet to determine precisely) perspectives

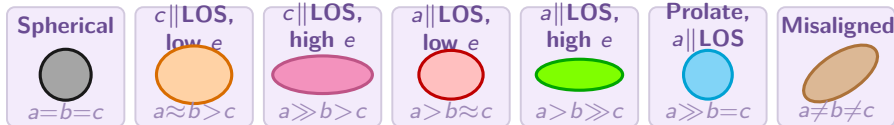
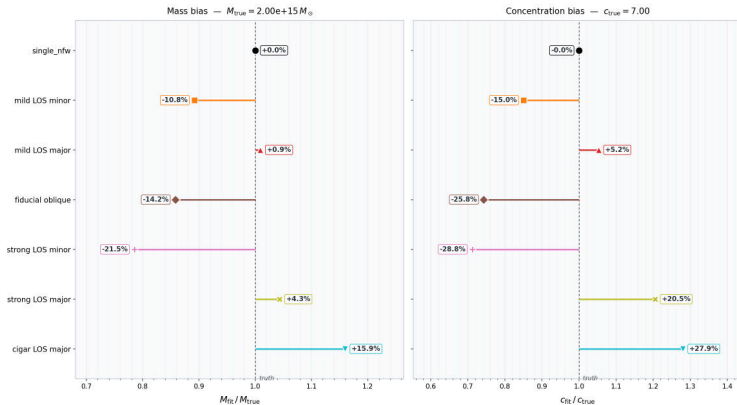


Take home message

excalibur: a modular relativistic raytracing code to quantify lensing effects and systematics. Lens geometry can bias shear profiles by $\approx 20\%$

Appendix: Joint κ, γ fit biases

Bias on inferred parameters when fitting a spherical NFW to non-spherical profiles



Sky-plane projections ($\perp$ LOS)

Backup: Spherical NFW profile

3D density (Navarro, Frenk & White 1997)

$$\rho(r) = \frac{\rho_s}{\frac{r}{r_s} \left(1 + \frac{r}{r_s}\right)^2}$$

- ρ_s : characteristic density
- $r_s = R_{\text{vir}}/c$, concentration c
- slope -1 for $r \ll r_s$; slope -3 for $r \gg r_s$

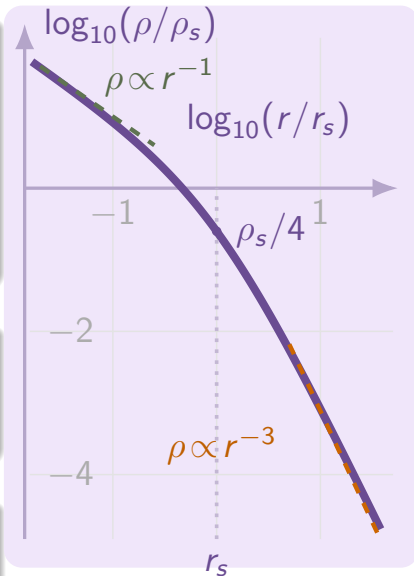
Projected convergence

$$\kappa(\xi) = \frac{2}{\Sigma_{\text{cr}}} \int_0^\infty \rho\left(\sqrt{\xi^2 + z^2}\right) dz$$

Analytic form: Wright & Brainerd (2000)

Role in this work

Reference (symmetric) case used to quantify triaxiality-induced biases.



Backup: Triaxial NFW profile

Triaxial generalization (Jing & Suto 2002)

Replace r by $m = \sqrt{x^2/a^2 + y^2/b^2 + z^2/c^2}$
($a \geq b \geq c$):

$$\rho_{\text{triax}}(\mathbf{x}) = \rho_{\text{NFW}}(m)$$

Projected convergence (LOS $\parallel c$)

$$\kappa(\xi_1, \xi_2) \propto \int_{-\infty}^{+\infty} \rho_{\text{NFW}}(m) dz$$

→ **Elliptical** contours, $q_{\perp} = b/a$.

Consequence

Fitting a spherical model to a triaxial halo biases the inferred mass and concentration (Corless & King 2007).

